HMCAS CPG v2.5 → v2.6: Summary of Changes

New CPGs

CPG 10.13 Vehicle Extrication & Care of the Trapped Patient (p.200)

- Self-extrication (or minimal help) is now the first-line approach. This includes children who can follow instructions, and patients with neck or back pain, soft neuro signs or central cord syndrome.
- Self-extrication is not suitable if the patient can't follow instructions or can't stand on at least one leg.
- Adds the "U-STEP Out" self-extrication tool.
- Conscious patients with no neuro deficit don't need manual in-line stabilisation while in the vehicle. Patients with hard neuro signs get a rapid extrication with gentle handling.
- While the patient is trapped, keep monitoring to a minimum. Only critical interventions: haemorrhage control, TXA, and analgesia (Penthrox or ketamine) to help extrication.
- A rigid collar can be used as an extrication device, then consider removing it. Use an "extrication buddy" and coordinate with Civil Defence.

CPG 12.13 Care of the Newborn

- Covers the main risks: hypoxia, hypoglycaemia, jaundice, sepsis, preterm birth and congenital abnormalities. For exposed organs, cover with cling film but don't wrap it all the way around.
- Normal ranges: RR 40–60, HR 100–160, temperature 36–37.5°C.
- Thermal care is mandatory: hat, wrap, keep temperature at or above 36.5°C. Preterm babies (<37 weeks) also need a thermal mattress and plastic bag wrap.
- Leave the cord intact for at least 60 seconds, then clamp about 5 cm from the umbilicus. Don't delay clamping if resuscitation is needed.
- Babies under 20 weeks are not viable and don't need resuscitation. Includes guidance on false signs of life and supporting the parents. If in doubt, resuscitate.
- Hypoglycaemia (RBS <2.6 mmol/L): give buccal glucose. No glucagon, and IV dextrose is generally not needed.
- Prenotify the receiving obstetric unit if the baby has complications or is being resuscitated.

Cardiac arrest (CPG 2.1–2.4)

All four cardiac arrest flowcharts were redesigned. Drug timing and defibrillation energy changed in both adults and children.

Adult (medical and trauma)

Item	v2.5	v2.6
Adrenaline, non-shockable	After the SGA cycle, once IV/IO is in	As soon as possible once IV/IO is in, then every 4 min
Adrenaline, shockable	After the 2nd shock (flowchart)	After the 3rd shock, then every 4 min
Amiodarone	300 mg after 3rd shock, 150 mg after 4th	300 mg after 3rd shock, 150 mg after 5th
Zoll energy	120J, 150J, 200J	150J, 200J, 200J (LP15 unchanged)
DSED	All later shocks DSED	Max 3 DSEDs, then standard anterior-posterior pads at max joules

Item	v2.5	v2.6
Vector change	APs	APs and CCPs; pad placement diagrams added
Mechanical ventilator	Not recommended	CCPs may use it (settings below)
Gastric tube via SGA	CCPs	APs and CCPs

Other adult changes:

- **CPR targets:** rate 100–120/min, depth 5–6 cm, pauses under 5 seconds before and after each shock. New rule: "if in doubt, assume arrest". Telephone CPR added to the chain of survival.
- **Capnography:** waveform capnography is mandatory with any advanced airway; aim for $EtCO_2$ of about 25 mmHg.
- **Ventilator settings:** volume controlled or pressure regulated, tidal volume 6–8 ml/kg, FiO_2 1.0, rate 10, I:E 1–1.5, PEEP 0–5, Pmax 60–70, trigger off. Go back to BVM if it isn't effective.
- **Other:** no neuromuscular blockers for CPR-induced consciousness. Check e-CPR criteria (CPG 2.9). "Update NCC" added to the flowchart.
- **Trauma arrest:** uses HOTT (hypovolaemia, oxygenation, tamponade, tension pneumothorax). Keep scene time to a minimum. Give adrenaline after 500 ml of fluid, then every 4 minutes.

Paediatric

Item	v2.5	v2.6
Start of CPR	CPR, BVM when ready	5 ventilations, then CPR at 15:2 (was 15:2 or 30:2)
Pad position	Anterior-lateral; anterior-posterior for vector change	Anterior-posterior first choice under 25 kg; either above 25 kg
CCP energy escalation	After 3 shocks, max 10 J/kg	After 5 shocks, in 2J steps, max 8 J/kg
Adrenaline, shockable	After the 2nd shock (flowchart)	After 4 min of CPR, 0.01 mg/kg (max 1 mg), every 4 min
Amiodarone	5 mg/kg	5 mg/kg after 3rd shock (max 300 mg) and after 5th (max 150 mg)
Fluids	10–20 ml/kg × 3 (medical); 10 ml/kg × 2 (trauma)	10 ml/kg × 3
Dextrose 10%	2.5 ml/kg	2 ml/kg (0.2 g/kg)
ETI after ~10 min	CCPs must consider	CCPs may consider
Needle decompression (trauma)	CCP or Delta	CCP only
Trauma flowchart outcome	Option to consider terminating	Continue resuscitation and transport

Other paediatric changes:

- Start CPR if HR is under 60 with signs of poor perfusion (pallor, mottling, central cyanosis).
- In non-shockable rhythms, give adrenaline as soon as possible.
- Ventilation rates by age: under 1 year 25, 1–8 years 20, 8–12 years 15, over 12 years 10 per minute.
- Two-thumb encircling technique is required for infants under 1 year.
- Calculate doses on actual weight where possible.
- **CPG 1.2 weight table:** paediatric weight-for-age values revised. For example, term to 2 months is now 4–5 kg (was 3–4 kg); 14 years and over is now >49 kg (was 46–50 kg).

Newborn resuscitation (CPG 2.5)

Ventilation now starts with 5 inflation breaths and a PEEP valve, replacing PPV without PEEP.

- **Assessment:** structured into breathing, heart rate, response to tactile stimulation, and tone. Skin colour is no longer used to judge oxygenation.
- **Airway:** jaw thrust is preferred. Suction only for obstruction, under direct vision (the meconium indication was removed). Guidance added on when to use an OPA and an SGA.
- **Ventilation:** 5 inflation breaths of 2–3 seconds, ideally within 60 seconds. PEEP 6 cmH_2O: room air at 32 weeks or more, 30% oxygen under 32 weeks.
- **Compressions:** if HR stays under 60 after 30 seconds of ventilation. 3:1 with 100% oxygen, two-thumb technique, reassess HR and breathing every 30 seconds.
- **Targets:** SpO_2 above 80% and HR above 100 by 5 minutes. Temperature 36.5–37.5°C.
- **Fluids and glucose:** 10 ml/kg normal saline, can be repeated once. Dextrose 10% 2 ml/kg if hypoglycaemic.
- **Defib (rare):** 4 J/kg with pads anterior-posterior.
- **Cord:** delay clamping if possible, but clamp and cut straight away if urgent resuscitation is needed.
- **Parents:** may stay during resuscitation; assign a paramedic to support them if resources allow.

Post-ROSC care and special circumstances (CPG 2.6, 2.8)

Post-ROSC care adds ice-pack cooling for comatose adults and a tighter SpO_2 target.

- **Oxygen:** SpO_2 94–98%; avoid hyperoxaemia (was SpO_2 above 90% or 94%).
- **Cooling (new):** ice packs to the axillae and groin for comatose adults after a medical arrest. Don't let temperature drop below 33°C, and tell the hospital.
- **Paediatric ROSC fluid:** 10 ml/kg boluses (was 20 ml/kg).
- **Dextrose 10%:** 2 ml/kg (was 2.5 ml/kg).
- **New sections in 2.8:** hyperthermia (actively cool to below 39°C), drowning (5 initial breaths with BVM and 100% oxygen, consider e-CPR), and cardiac arrest during dialysis.
- **Hypothermic arrest:** transport all of these patients without signs of undeniable death (was "consider") and consider e-CPR. Give adrenaline every 6–10 minutes once temperature is 30–35°C.
- No neuromuscular blockers for CPR-induced consciousness.

Stroke (CPG 3.1)

The 15-hour cut-off is gone: destination now depends on onset time and, for 8–24 hours, a RACE score. Previously all suspected strokes went to HGH ED (P1 if under 15 hours, P2 if over).

Onset / last known well	Action	Destination
Wake-up or unknown onset	Prenotify via CC:E	P1 to HGH ED
8 hours or less	Prenotify via CC:E	P1 to HGH ED
More than 8 and less than 24 hours, RACE 5 or more	Prenotify via CC:E	P1 to HGH ED
More than 8 and less than 24 hours, RACE under 5	—	P2 to nearest appropriate ED
24 hours or more	—	P2 to nearest appropriate ED

- **RACE scale added** (0–9) to screen BEFAST-positive patients for large vessel occlusion. A score of 5 or more suggests LVO.

- **Prenotification checklist:** age, time last seen well, BEFAST, RACE (if done), RBS, BP, anticoagulant status, ETA.
- **Management:** Penthrox removed; paracetamol and an antiemetic if required; oxygen only as required.

Trauma

Paediatric TXA is now an AP-level intervention, and antepartum haemorrhage is no longer a TXA indication.

- **10.3 Spinal: new section on cord injury and neurogenic shock.**
  - Consider early intubation for C1–4 injuries with airway or breathing compromise, keeping in-line stabilisation.
  - MAP 85–90 mmHg for isolated neurogenic shock, or 65 mmHg or more if combined with haemorrhagic shock.
  - Noradrenaline is the preferred vasopressor; add an adrenaline infusion if bradycardia and hypotension persist.
  - Revised spinal motion restriction (SMR): consider removing the combi-carrier once the patient is on the stretcher (keep it for unconscious patients), head blocks always, stretcher can be raised up to 30°.
- **TXA, paediatric:** 15 mg/kg (max 1 g) moves from CCP to AP level in the haemorrhage, TBI, chest and pelvic charts. Paediatric TBI with GCS 12 or less is a new indication.
- **TXA, obstetric:** antepartum haemorrhage removed. TXA deleted from ectopic pregnancy, miscarriage, abruption and praevia; it stays for postpartum haemorrhage only.
- **10.10 Trauma bypass:** abdominal “seat belt sign” added as a criterion.

Smaller changes

- **4.3 Broad complex tachycardia:** adult synchronised cardioversion 150J (was 200J).
- **6.4 Nausea and vomiting, and Ondansetron formulary:** CCPs may give an antiemetic and then hand a stable patient to APs for transport.